

The Neuroscience of Translation

Maria Tymoczko

University of Massachusetts

The neurological mechanisms involved in translating and interpreting are one of the chief known unknowns in translation studies. Translation studies has explored many facets of the processes and products of translation and interpreting, ranging from the linguistic aspects to the textual aspects, from the politics of translation to implications from cognitive science, but little is known about the production and reception of translation at the level of the individual brain and the level of molecular biology. Much of this terra incognita will be explored and illuminated by neuroscience in the coming quarter century, and significant discoveries pertaining to language processing in translation will be made during the coming decade, linking observable behaviors at the macro-level with knowledge of what happens in the production and reception of translation at the micro level of the neuron and the neuronal pathways of the brain. In the past two decades powerful new techniques for observing brain function in healthy living individuals have been devised. To a large extent neuroscience has become a rapidly developing field because of new technologies that make it possible to monitor the brain as it actually works, to document neural pathways, and even to track the activity of specific neurons. This article focuses on discoveries in neuroscience pertaining to perception, memory, and brain plasticity that have already achieved consensus in the field and that have durable implications for the ways we will think about translation in the future.